

Module 03: SQL Foundations, Data Types, and DDL Commands

SQL Command Ecosystem & DDL Structural Lifecycle

Characteristics, Data Types & DDL Commands: CREATE, ALTER, DROP, TRUNCATE

1. SQL Core Characteristics & Types

Essential SQL Data Types & Storage:

- CHAR(n): Fixed-length string. Space padded if shorter. Fast for fixed codes.
- VARCHAR(n) / VARCHAR2(n): Variable-length. Space efficient.
- INT, SMALLINT, BIGINT: Discrete integer values.
- DECIMAL(p, s) / NUMERIC: Fixed-point precision p with s decimals (currency).
- DATE, TIME, TIMESTAMP: Calendar date & fractional second timestamps.
- BLOB, CLOB: Binary & Character Large Objects (Images, PDFs, JSON).

The 4 SQL Command Families:

- DDL (Data Definition): CREATE, ALTER, DROP, TRUNCATE, RENAME.
- DML (Data Manipulation): SELECT, INSERT, UPDATE, DELETE.
- DCL (Data Control): GRANT, REVOKE (Permissions & security).
- TCL (Transaction Control): COMMIT, ROLLBACK, SAVEPOINT.

DDL operations are **AUTO-COMMITTED!** Cannot be rolled back!

2. DDL Structural Commands Deep Dive

CREATE TABLE

```
CREATE TABLE Student (Roll INT PRIMARY KEY, Name VARCHAR(50));
```

Allocates data dictionary metadata and initial disk extent.

ALTER TABLE (Structural Modifications)

- ADD: ALTER TABLE Student ADD Email VARCHAR(100);
- DROP: ALTER TABLE Student DROP COLUMN Email;
- MODIFY: ALTER TABLE Student MODIFY Name VARCHAR(100);
- RENAME: ALTER TABLE Student RENAME COLUMN Roll TO RollNo;

TRUNCATE vs DROP vs DELETE (AKTU Comparison)

- DROP: Deletes table DATA + SCHEMA completely from disk. Free space.
- TRUNCATE: Deletes ALL rows. SCHEMA remains. High speed (resets HWM).
- DELETE: DML command. Deletes specific rows. Generates undo logs.

TRUNCATE is DDL (Auto-commit, no WHERE clause, lightning fast!).

DDL Auto-Commit: DDL commands (CREATE, ALTER, DROP, TRUNCATE) modify the data dictionary directly and are automatically committed!

1. Essential SQL Data Types

Type	Storage	Description
CHAR(n)	Fixed n bytes	Fixed length string padded with trailing spaces. Ideal for fixed-width identifiers.
VARCHAR(n)	Variable length	Dynamic string up to n characters. Space-efficient for names and descriptions.
DECIMAL(p, s)	Exact precision	Fixed-point numeric with p total digits and s fractional digits (e.g. monetary amounts).
TIMESTAMP	Date + Time	Calendar date, hour, minute, second, and fractional microsecond values.

2. TRUNCATE vs DROP vs DELETE

Feature	DROP	TRUNCATE	DELETE
Category	DDL	DDL	DML
Schema State	Completely Removed	Preserved	Preserved
Rollback	No	No	Yes (via ROLLBACK)
WHERE Clause	No	No	Yes
Speed	Instant	Fast (Resets HWM)	Slower (Row logging)